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OVERVIEW

This procedure may be performed after installation of a standard Receiver, Multicoil Receiver, Receiver 0 for Fast Receiver, or a Fast Receiver Module to inspect BACC plots. BACC must be run before EPT.

Asymmetry of the receiver analog filter response leads to ghosting artifacts with EPI. If the receiver response is known, the ghosting artifact is easily removed. The Bandwidth Asymmetry Correction Characterization (BACC) test is a loopback characterization to measure the frequency response. Ghosting artifacts in the phase encode direction occur with both the Fast Receiver (FR) and Standard Receiver (SR), and in both single-shot and multishot EPI acquisitions. The BACC must characterize all SRs and FRs that reside on any EPI-compatible system.

1- SCAN PREPARATION

1. At the Operator Workspace, select the scan icon in the desktop control panel.
2. Click on **[New Pt]**, and enter the following:

   Id: **geservice <Enter>**
   Patient Name: **bacc <Enter>**
   Weight (Lb): **111 <Enter>**

3. Landmark on some point on the cradle (the scan uses TPS/ISE loopback so no RF is actually transmitted to the coil). At the keypad on the magnet front enclosure, press LANDMARK and ADV TO SCAN.

4. Set Patient Protocols to **Service**.

5. In the Protocol field, Type **o.42.1 <Enter>** (o=Other, 42=protocol number, 1=series number).

   OR
   Click on "Other" and select protocol **42** and series **1** from the menu.

6. Click on **[Accept]** to load the protocol.

7. Click on **[Save Series],** then **[Prepare to Scan]**.

8. Click on **[Research Operations]**, **[Display CVs]** with the right mouse button and enter the following settings:

   rhtype1 = **8 <Enter>** (default)
   cfrecvend = **0 <Enter>** (Non-multicoil) or
   3 <Enter> (Multicoil)
   fast_rec* = **0 <Enter>** (Normal Receiver(s)) or
   1 <Enter> (Fast Receiver if Available)
   Always set fast_rec = **0 for first pass*

   Note

   For Fast Receiver option, only. The BACC characterization must be run twice if the Fast Receiver option is available: once with “fast_rec = **0**” for standard receiver[s], and once with “fast_rec = **1**” for the Fast Receiver.

8. Click on **[Accept]**.

2- CORRECT CHARACTERIZATION

1. Click on **[Research Operations]**, **[Download]** with the right mouse button.
2. Click on **[Research Operations]**, **[Setup Params]** and set R1=**11**, R2=**15**, TG=**200**, then select **[Done]**.
3. Click on **[Manual Prescan]** to verify settings, then **[Done]**.
4. Select **[Scan]**. The scan takes about two seconds.
5. Wait for analysis screen to display the complete analysis. Once up, press the **<Q>** key or **<Enter>** on the keyboard to close the window.
6. If the Fast Receiver option is installed, run this scan again with the CV fast_rec=**1**. When the CV is changed, press **<Enter>** and then **[Accept]**.
7. Click on **[Research Operations]**, **[Download]**, then **[Scan**].
8. Wait for analysis screen to display the complete analysis. Once up, press the **<Q>** key or **<Enter>** on the keyboard to close the window.

3- DATA DISPLAY - CERD

From the Service Desktop, use the Report Manager to display BACC data files (these files have a ".BAT" file extension).

3-1 Header Data

The following is an example of BACC header data listing.

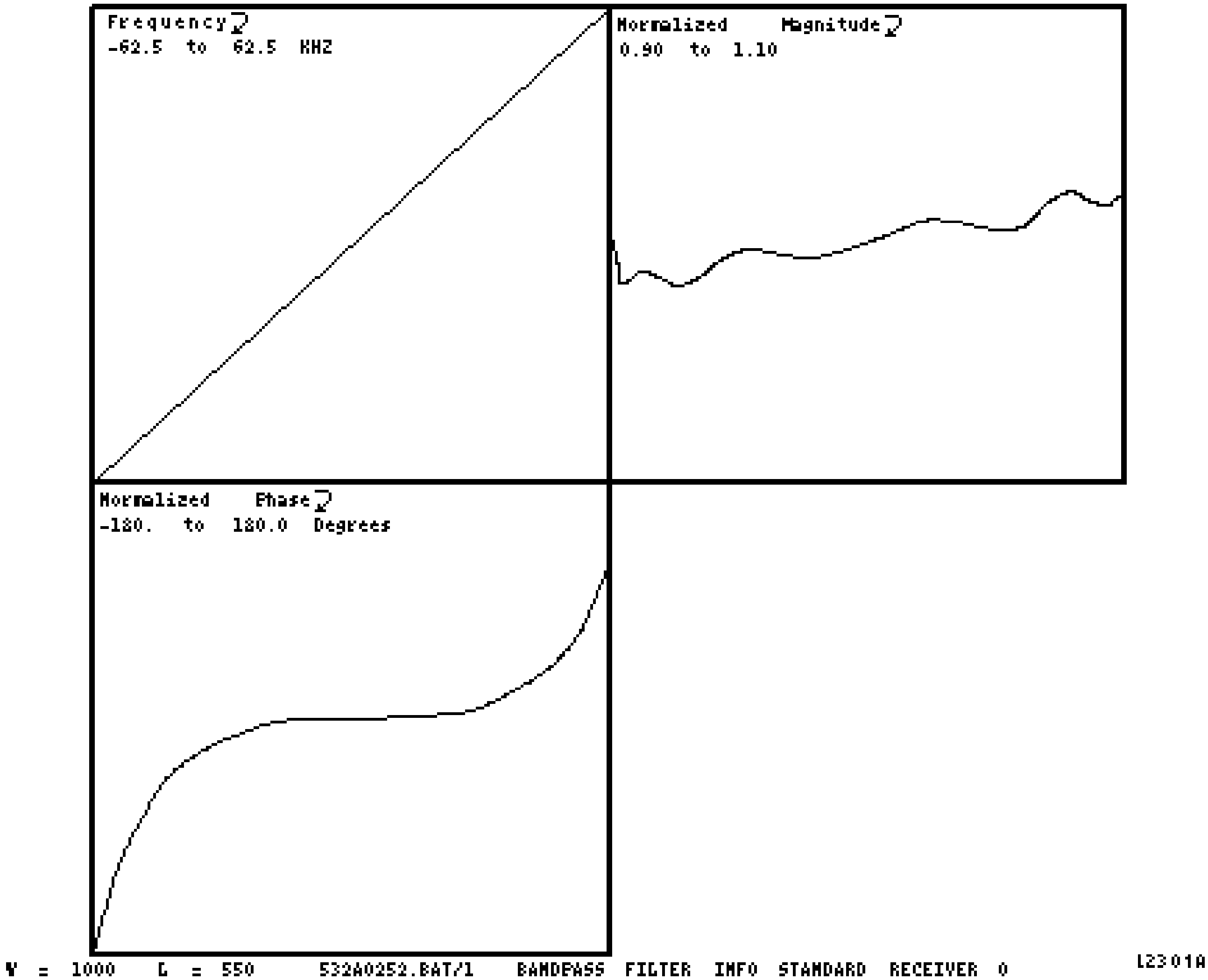
```
3A193912.BAT/0 Header Info RDF/GRP Revision: 1.0 /43
====================================================
SITENAME  = BAY 12
USN       = B15B0C0
MLN       = 9999
SRVCONFIG  = 09/27/95 15:28:55
EXCITER   = 000
RECEIVER  = S0/E0/PE0
XMTRFCOIL  = BODY
RCVRFCOIL  = BODY
FREQ      = 63866811 Hz
TIME      = 10/01/95 09:24:00
BASERUN   = 44032
CONFIGCODE = 001
SOFTREV   = 5.5
NUCLIDE   = 000
HEADERCODE = 0x00000000
RCVCOILGAIN = 1.880 R1 = 6 R2 = 14 TG = 135
------- Exam description -------
NO COMMENT
------- Series description -------

-----------------------------------------------------
Select (Help, Group #, Next, Files, Quit) [N]:
```

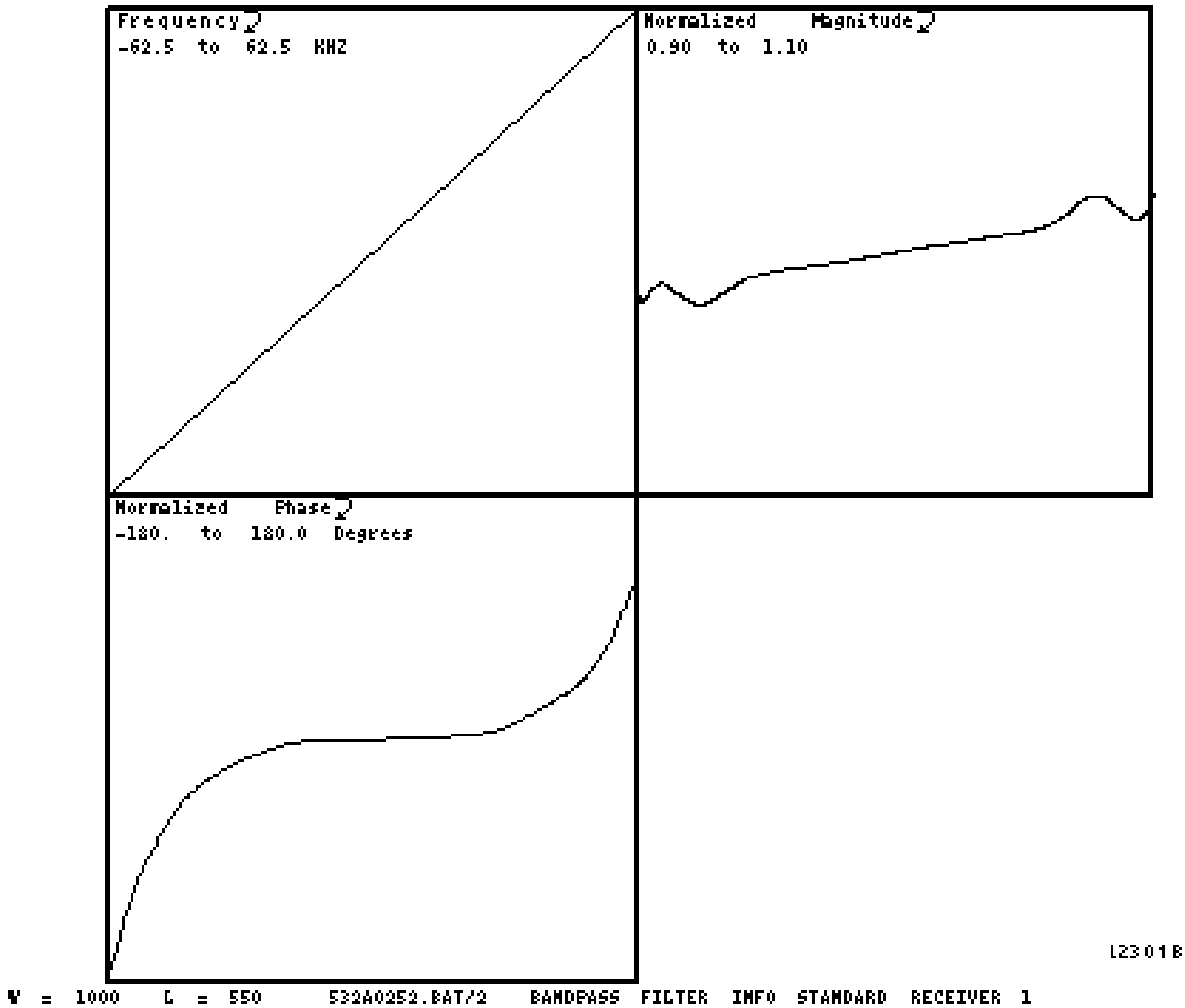
Enter desired selection.

3-2 Standard Receiver and Filter Displays - CERD

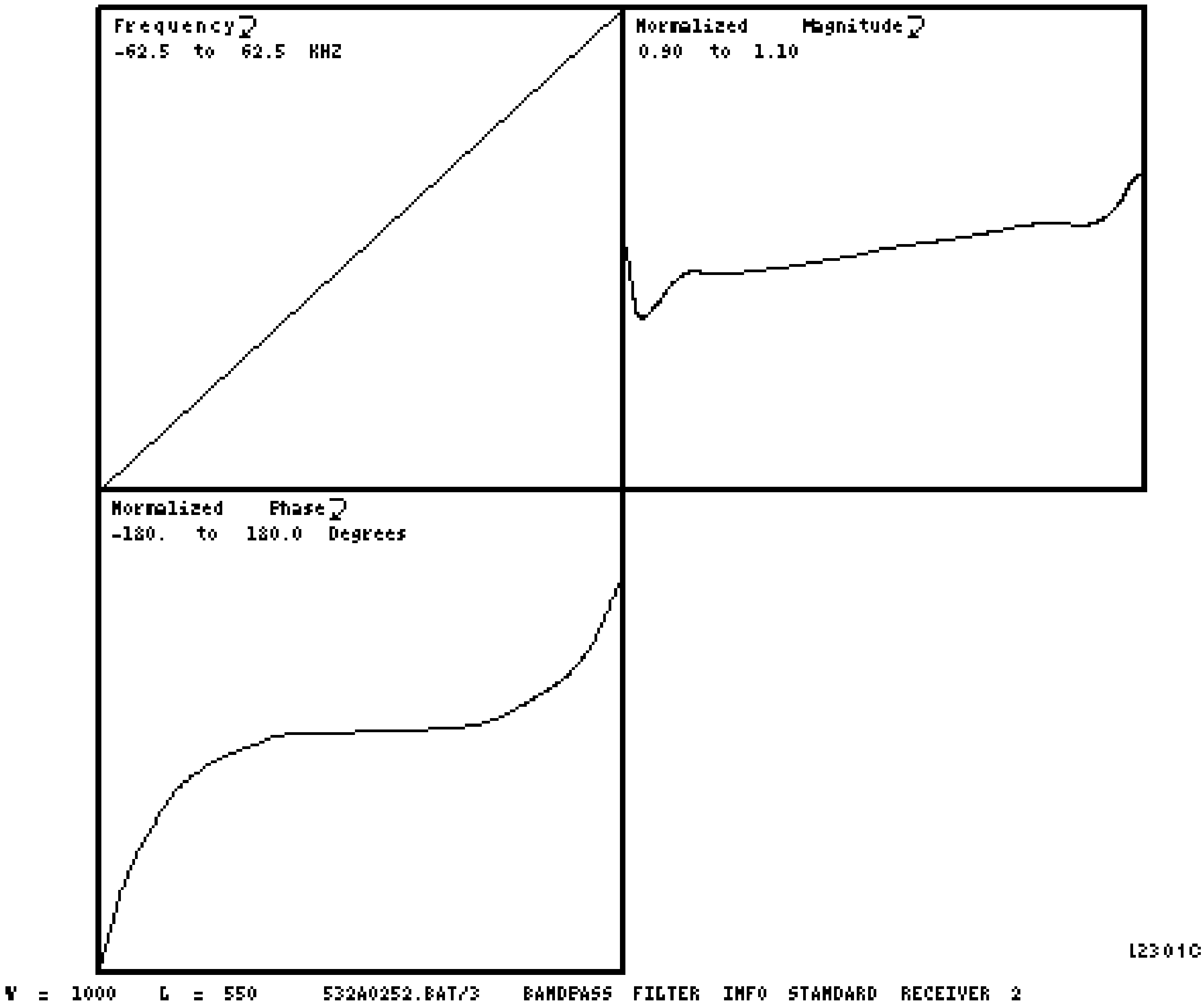
The following Illustrations are examples of plots for various receivers and filters:



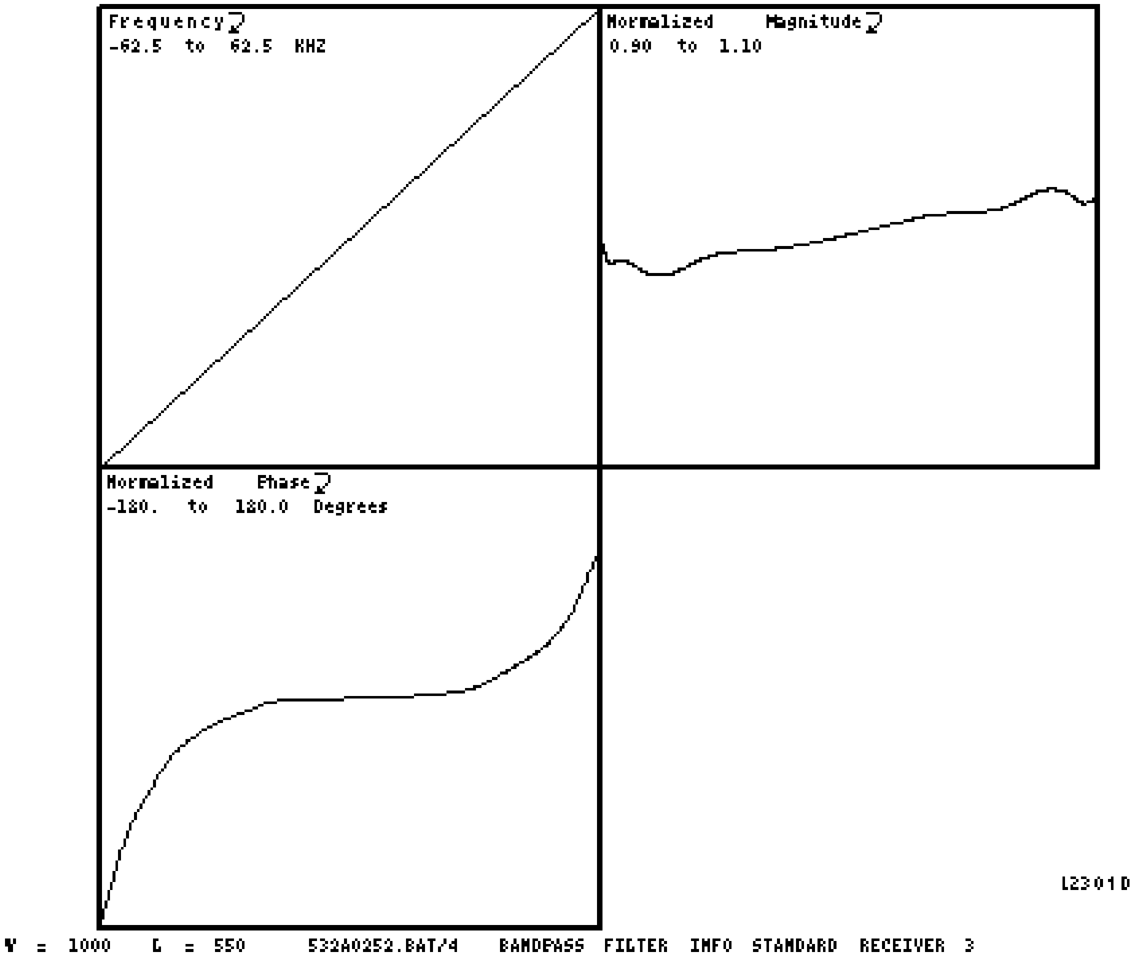
BACC RECEIVER 1 DISPLAY
ILLUSTRATION 3-1



BACC RECEIVER 2 DISPLAY
ILLUSTRATION 3-2



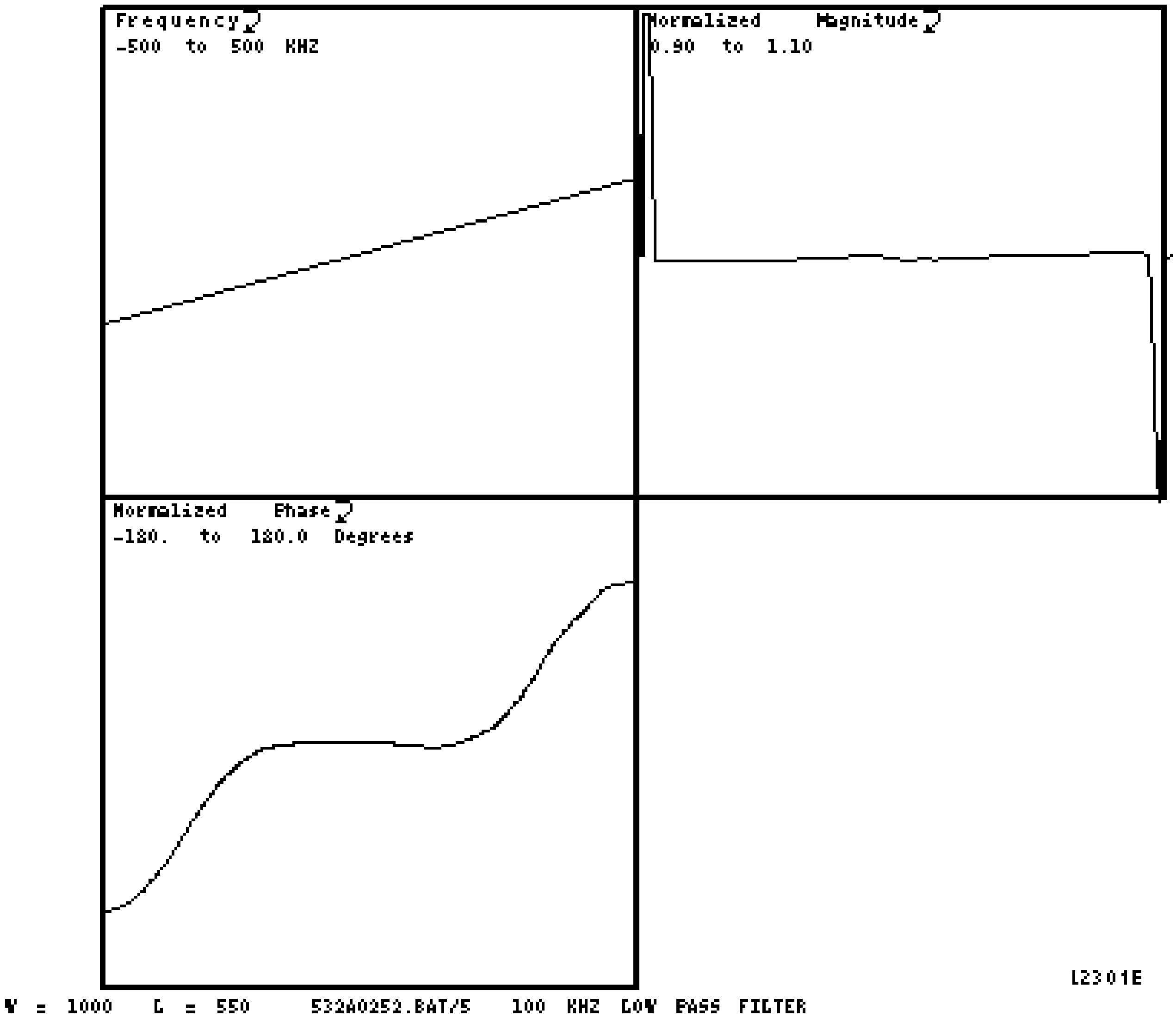
BACC RECEIVER 3 DISPLAY
ILLUSTRATION 3-3



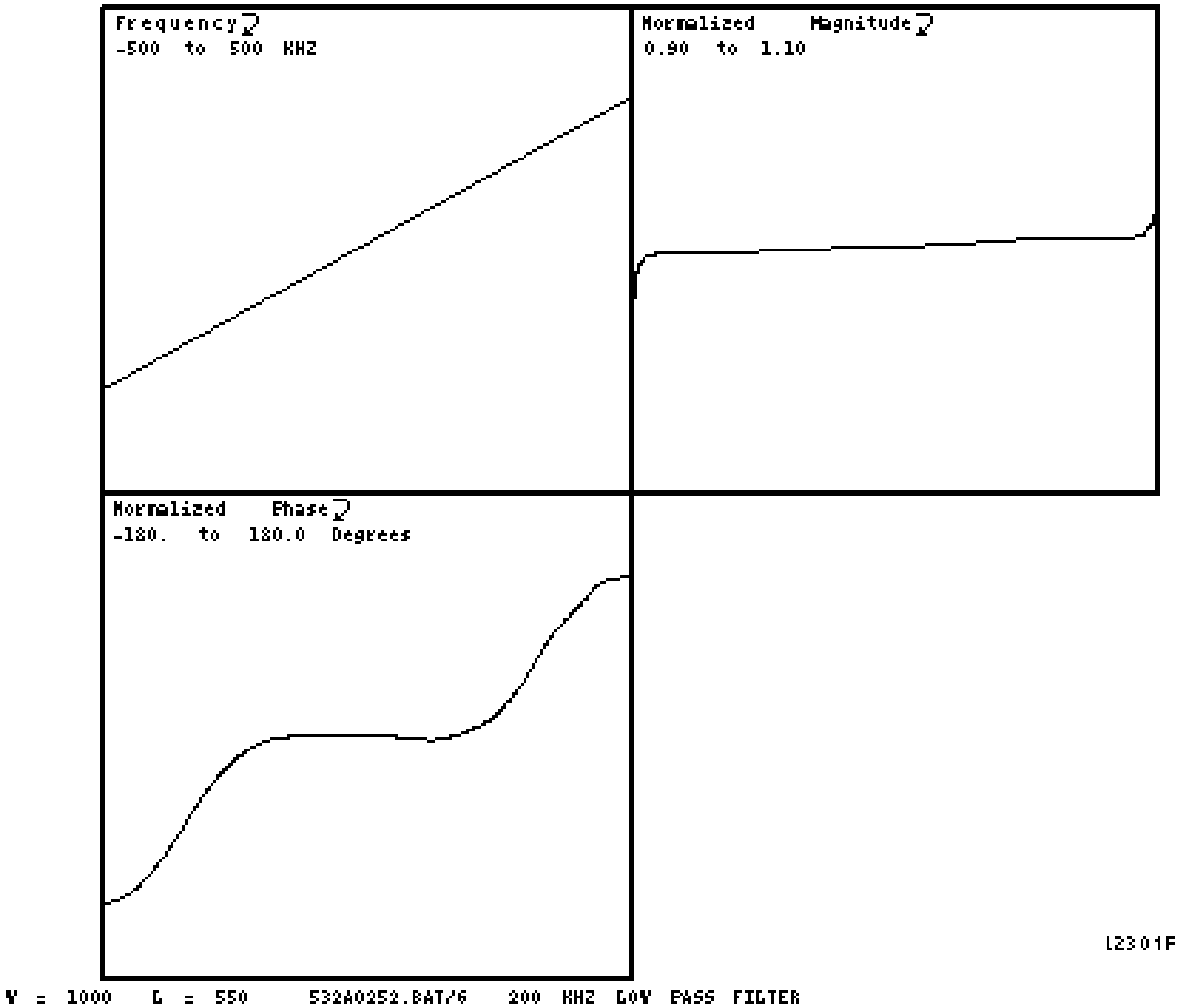
BACC RECEIVER 4 DISPLAY
ILLUSTRATION 3-4

3-3 Fast Receiver Filters and Displays - CERD

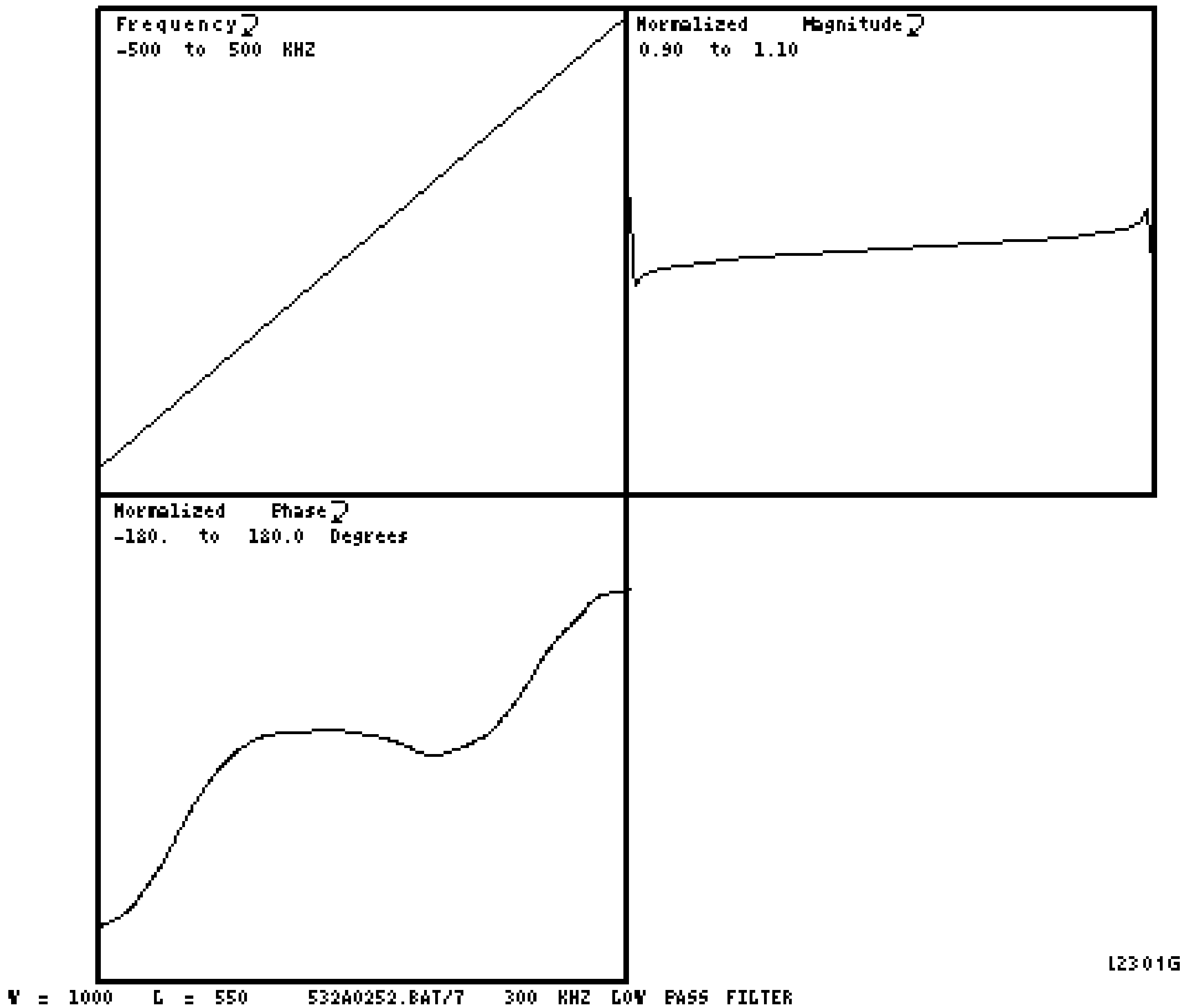
The following Illustrations are examples of plots for various receivers and filters:



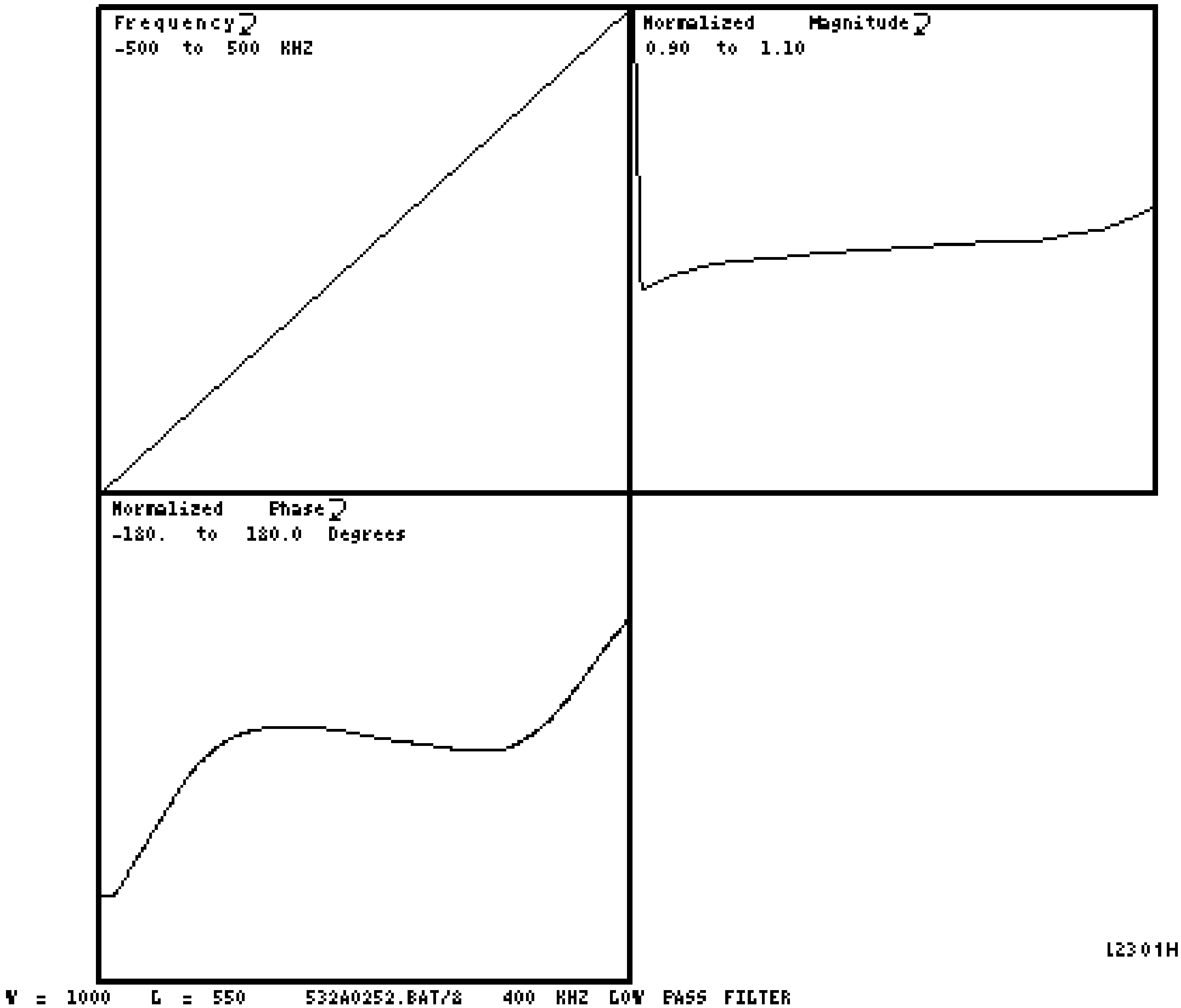
BACC FAST RECEIVER 100-KHZ FILTER DISPLAY
ILLUSTRATION 3-5



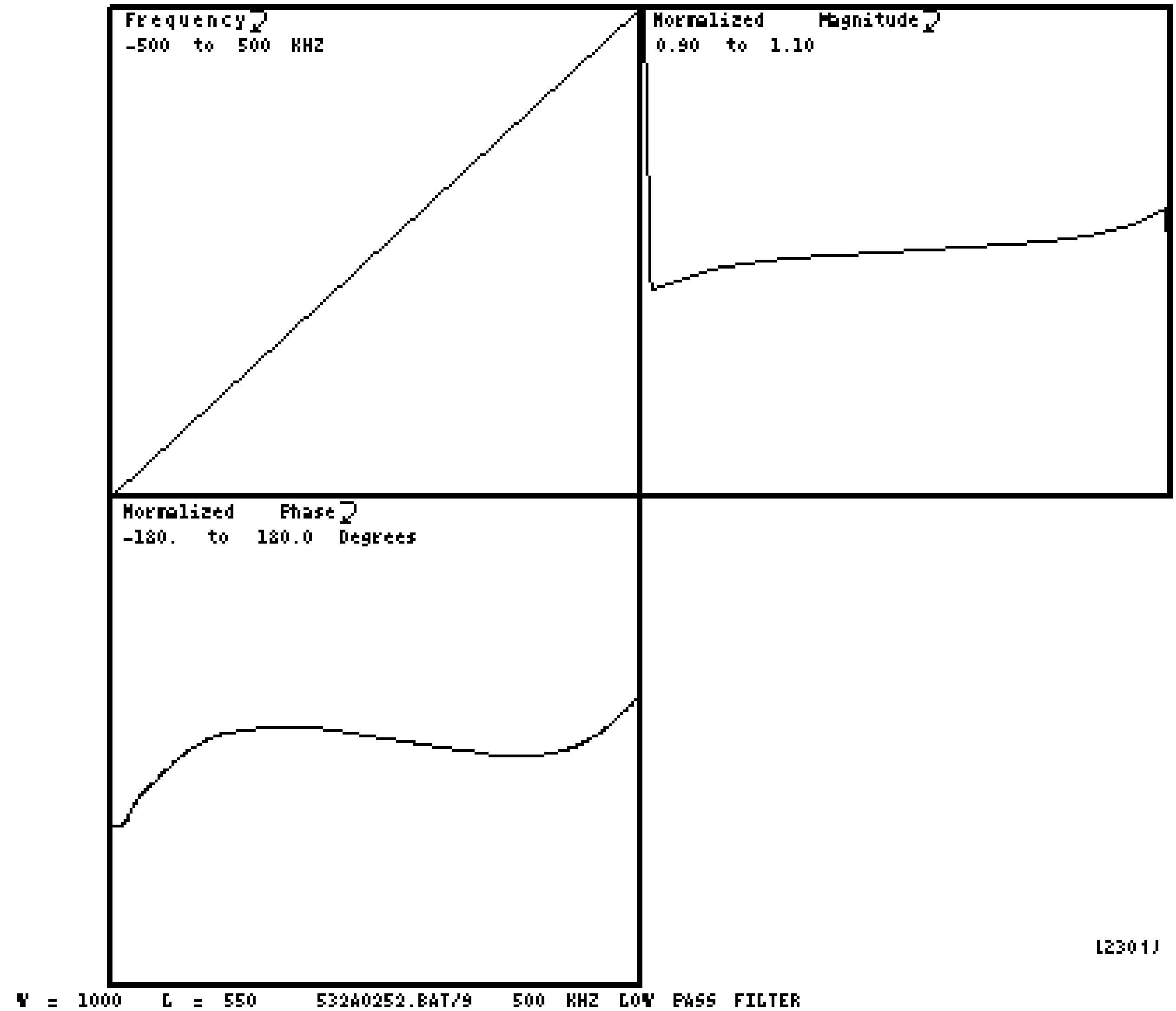
BACC FAST RECEIVER 200-KHZ FILTER DISPLAY
ILLUSTRATION 3-6



BACC FAST RECEIVER 300-KHZ FILTER DISPLAY
ILLUSTRATION 3-7



BACC FAST RECEIVER 400-KHZ FILTER DISPLAY
ILLUSTRATION 3-8



BACC FAST RECEIVER 500-KHZ FILTER DISPLAY
ILLUSTRATION 3-9

4 - DATA DISPLAY - UCERD

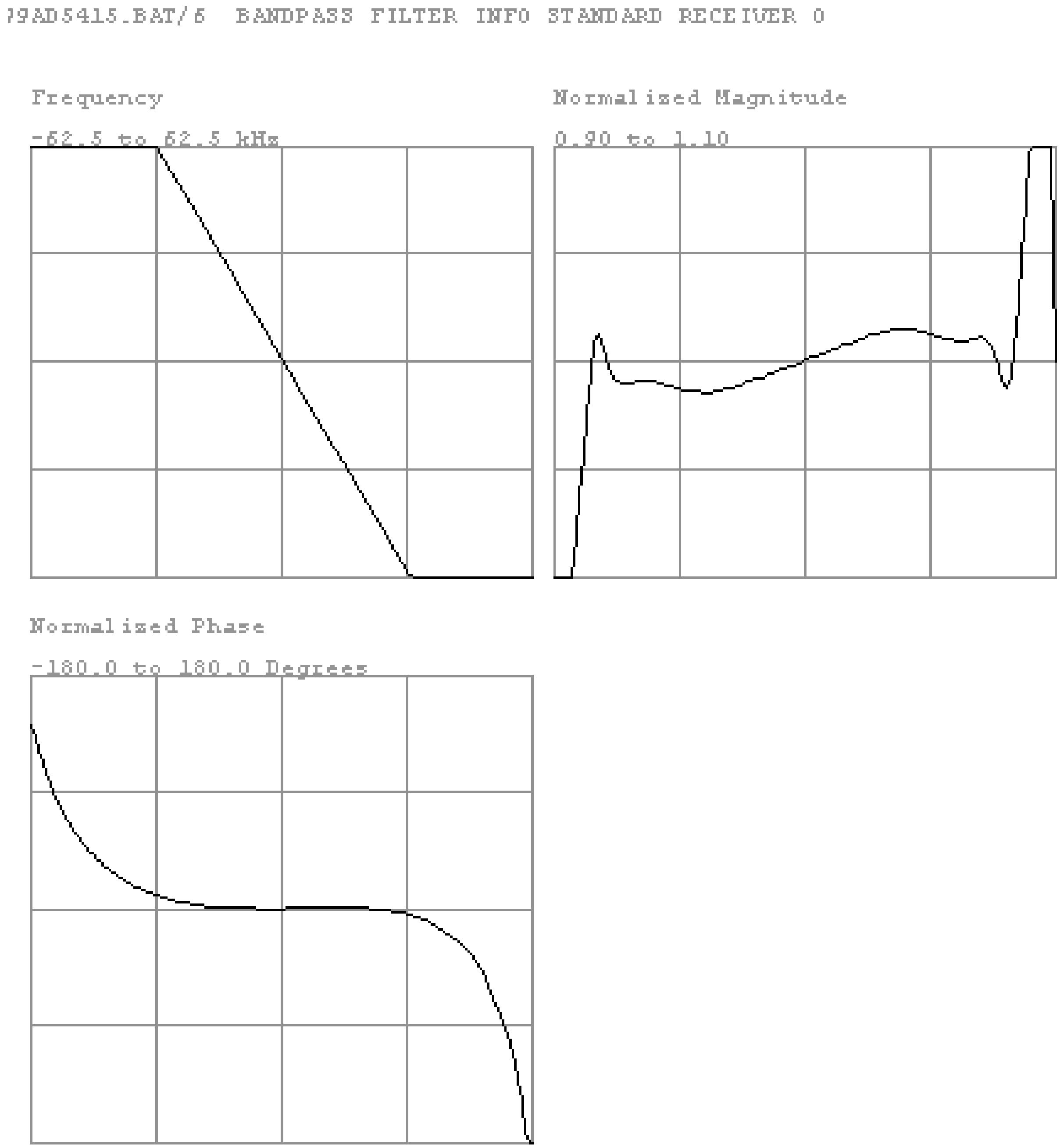
From the Service Desktop, use the Report Manager to display BACC data files (these files have a ".BAT" file extension). The UCERD files look different than the CERD files.

4-1 Header Display

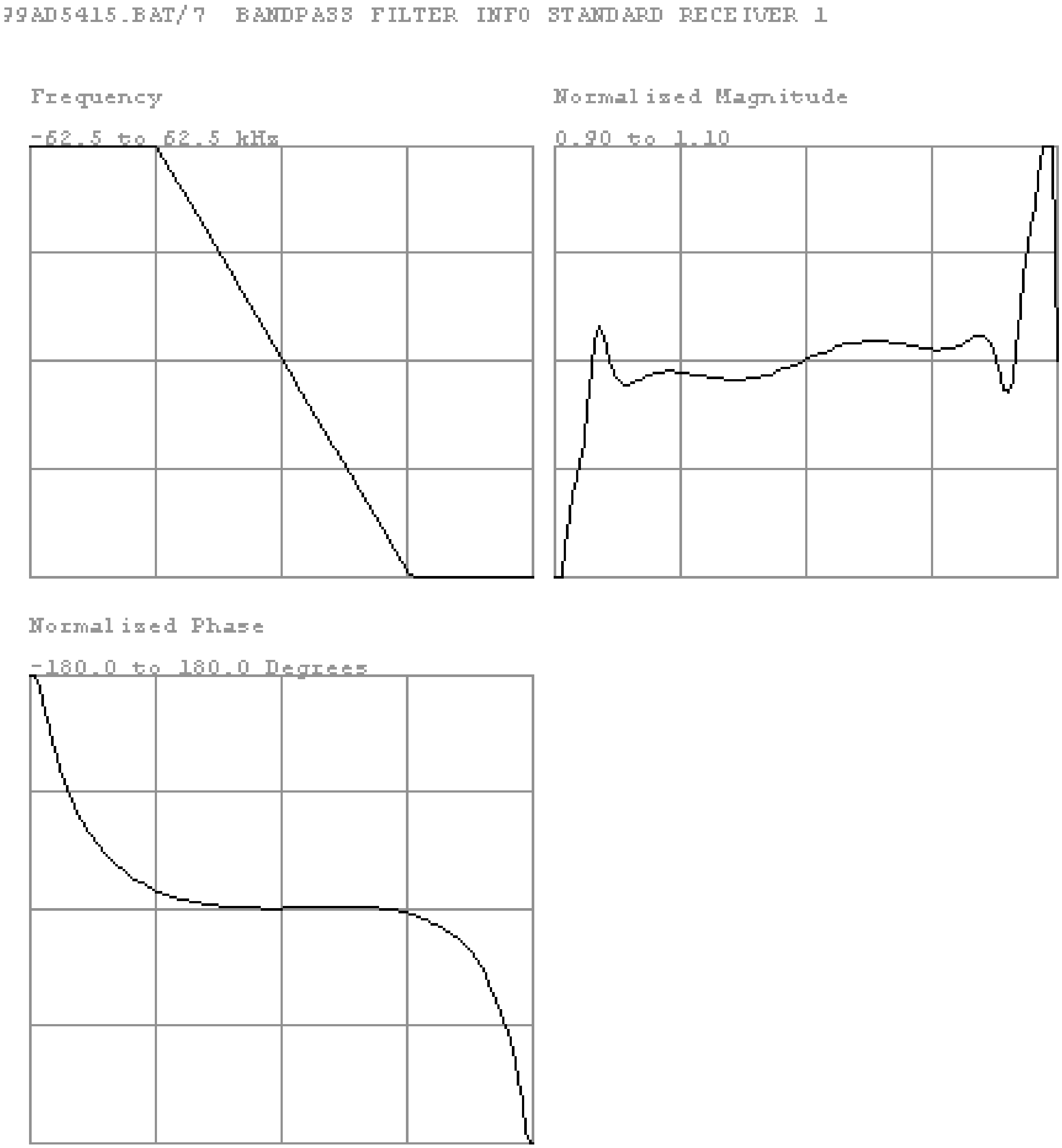
The Header display is the same for the UCERD and CERD. See Section 3-1.

4-2 Standard Receiver and Filter Displays - UCERD

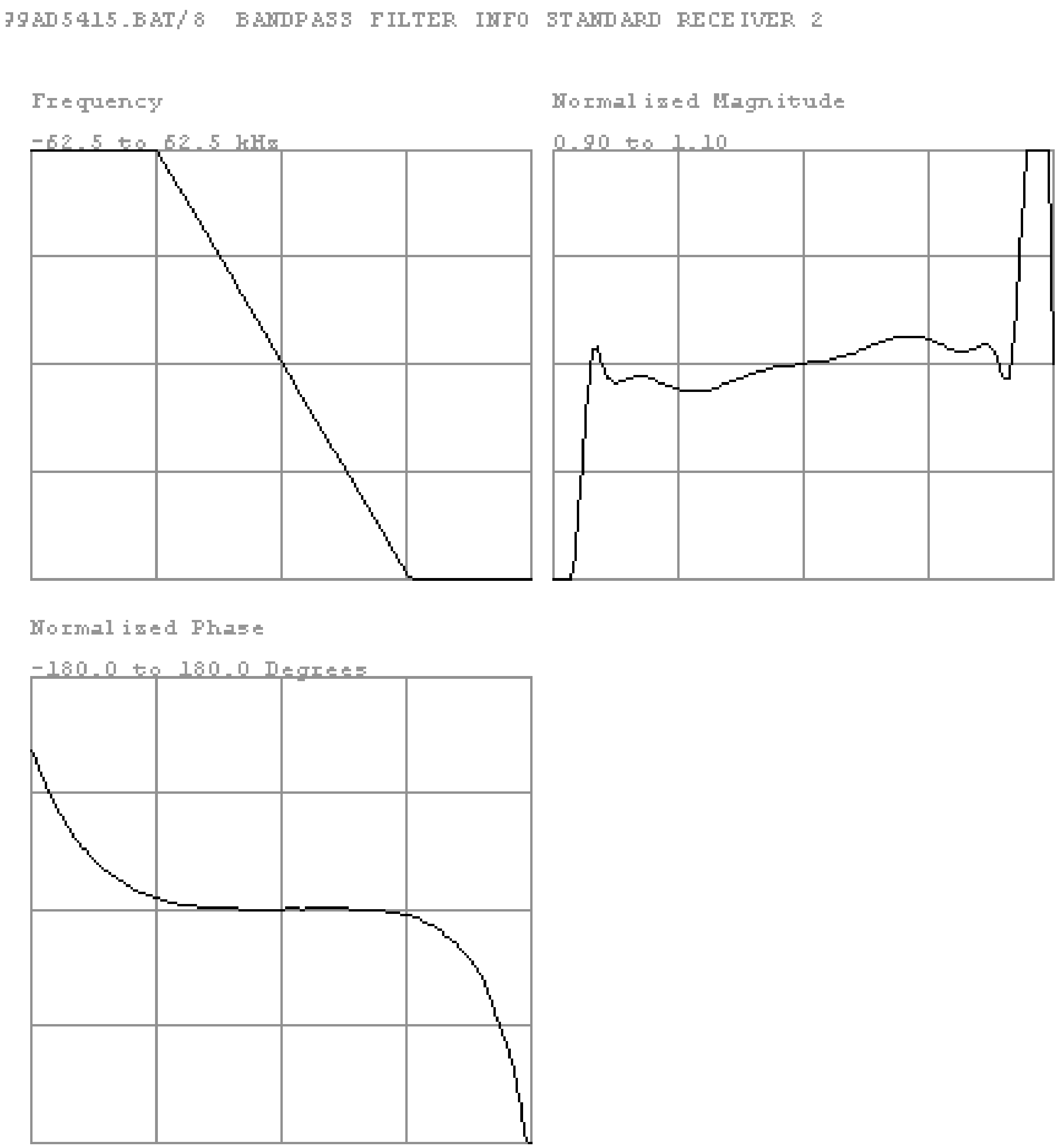
The following Illustrations are examples of plots for various receivers and filters:



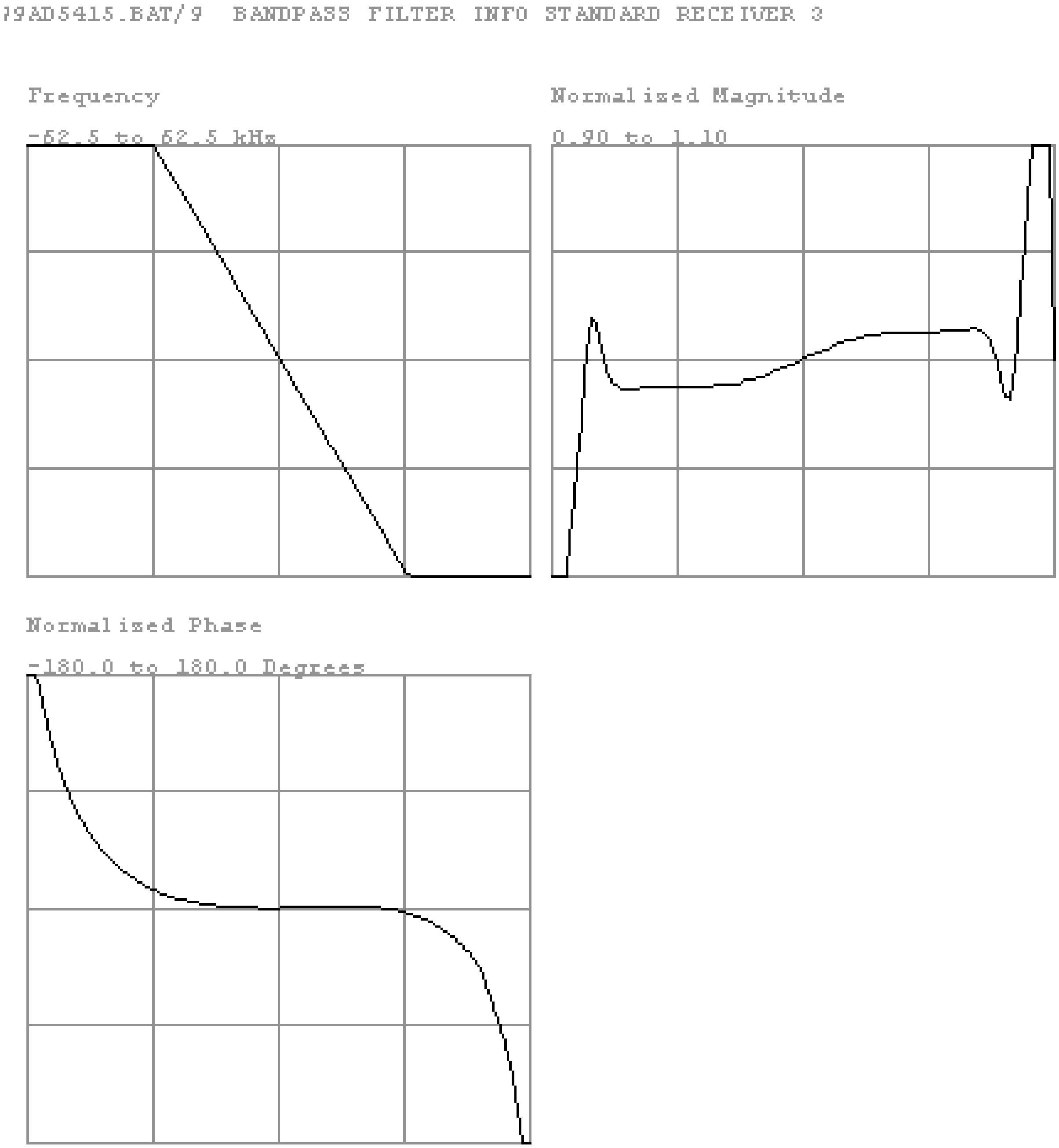
RECEIVER 0 - UCERD
ILLUSTRATION 4-1



RECEIVER 1 - UCERD
ILLUSTRATION 4-2



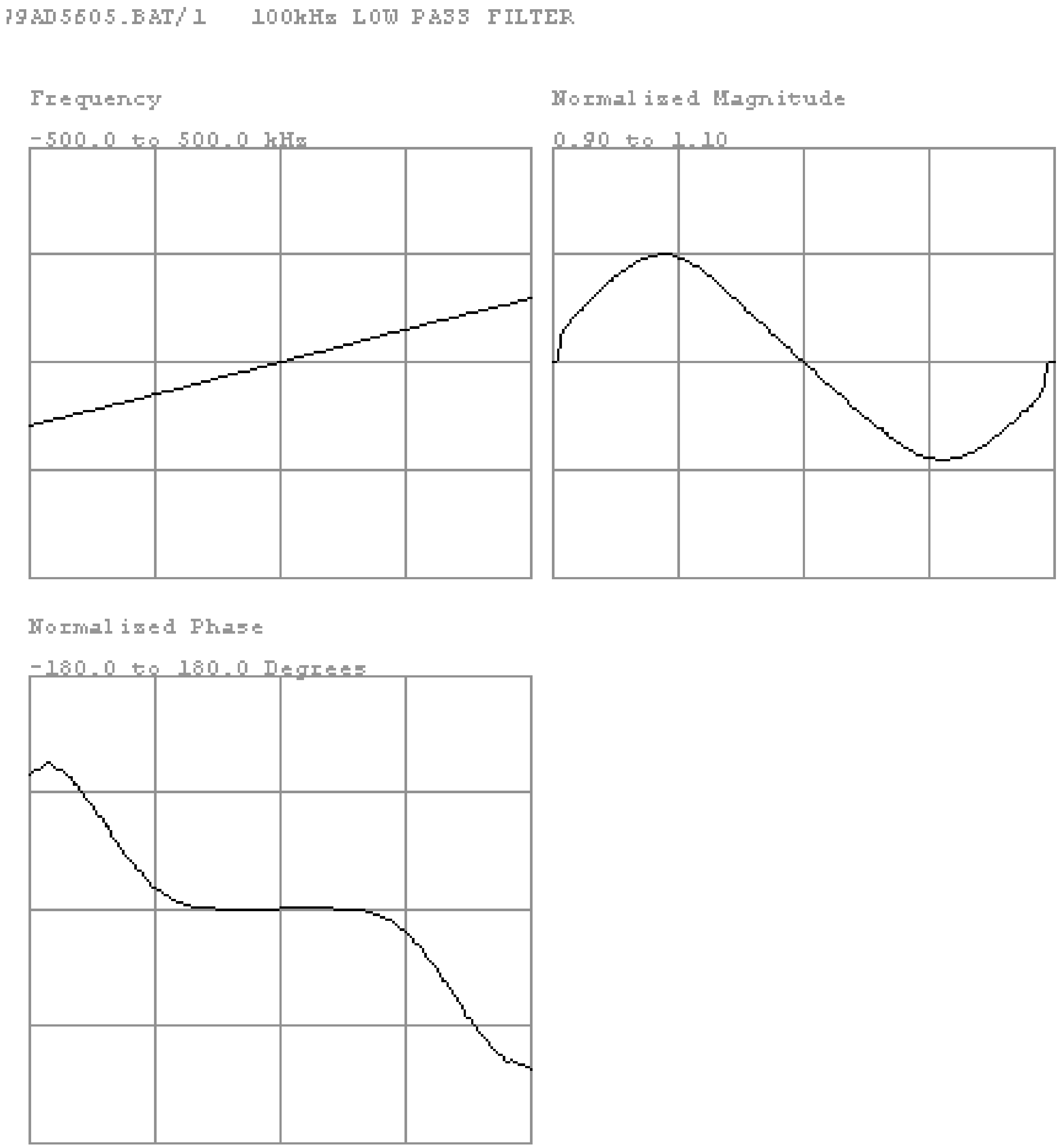
RECEIVER 2 - UCERD
ILLUSTRATION 4-3



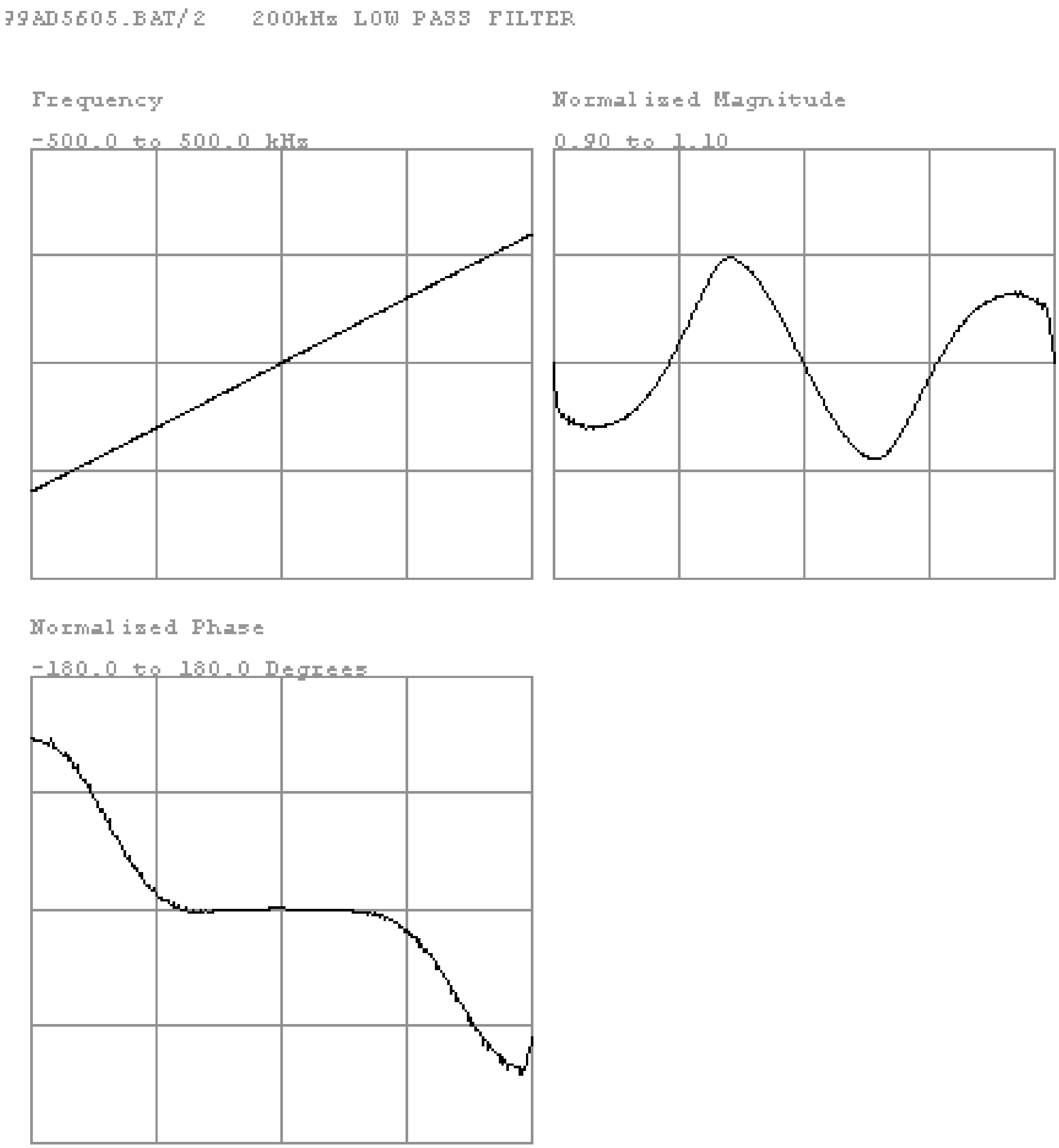
RECEIVER 3 - UCERD
ILLUSTRATION 4-4

4-3 Fast Receiver Filters and Displays - UCERD

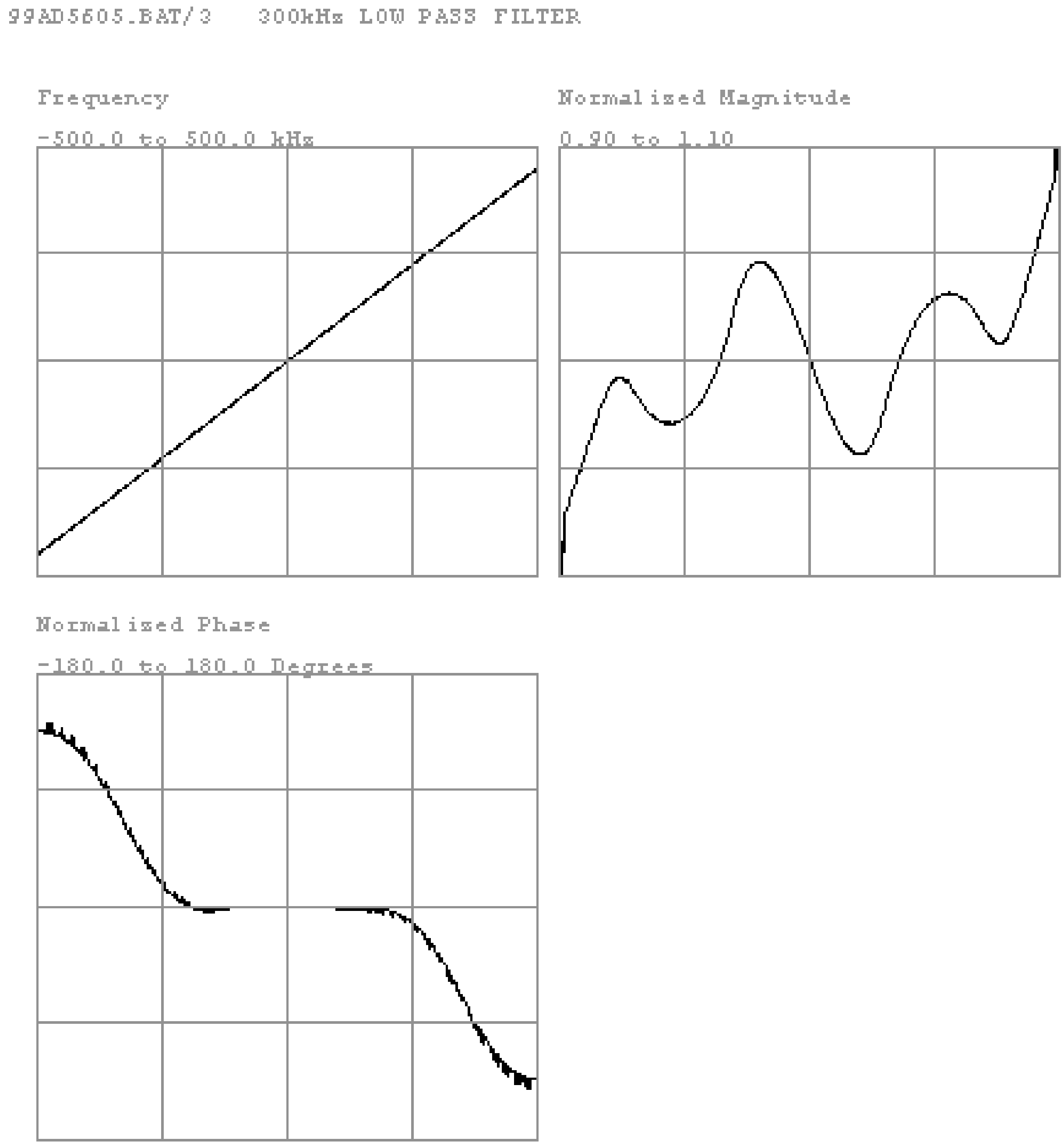
The following Illustrations are examples of plots for various receivers and filters:



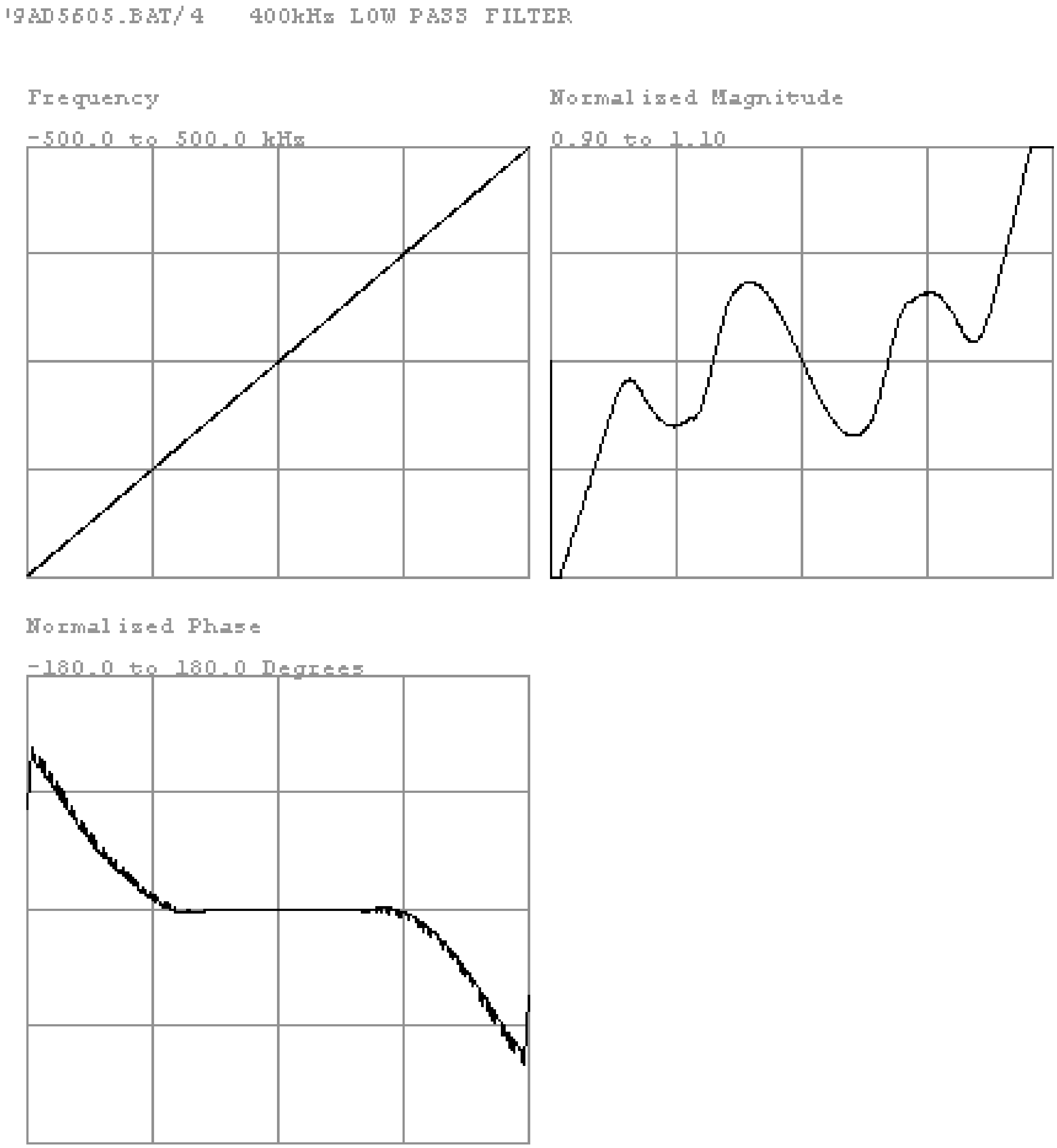
FAST RECEIVER 100 KHZ FILTER - UCERD
ILLUSTRATION 4-5



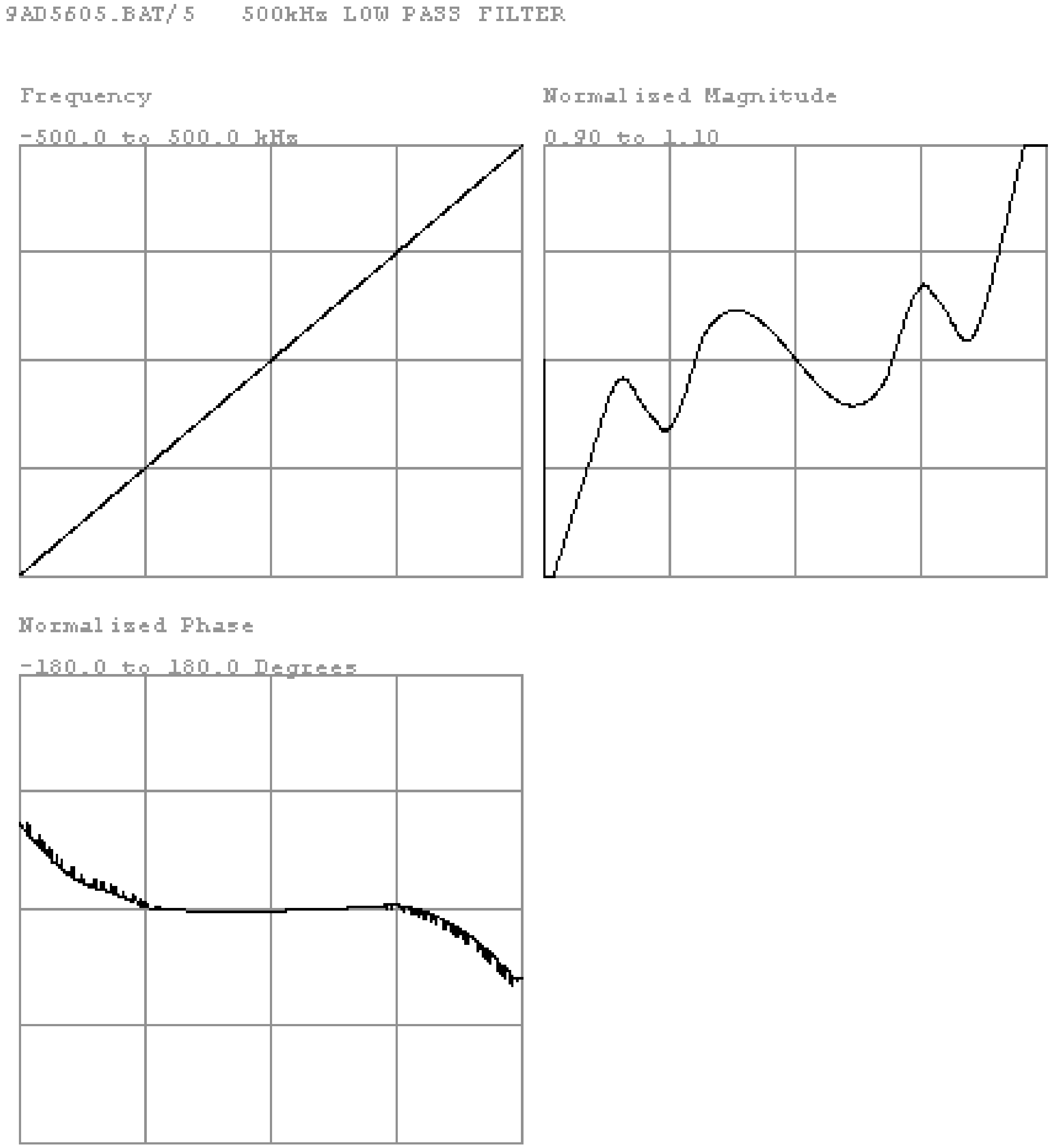
FAST RECEIVER 200 KHZ FILTER - UCERD
ILLUSTRATION 4-6



FAST RECEIVER 300 KHZ FILTER - UCERD
ILLUSTRATION 4-7



FAST RECEIVER 400 KHZ FILTER - UCERD
ILLUSTRATION 4-8



FAST RECEIVER 500 KHZ FILTER - UCERD
ILLUSTRATION 4-9

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 3, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.
1	Nov 2, 1998	M. Keber	Removed obsolete 8.1 information; misc. style guide cleanup.
2	Oct 13, 1999	M. Keber	Added correct proprietary heading to document.
3	Dec 26,1999	G. Boerner	Landmarking per SPR MRIge56065 and minor edits per 8.3 validation.
4	June 20, 2000	M. Jones	Minor clarity enhancements.
5	Dec 4, 2001	M. Keber	Removed Sec 1 references to proprietary and non-proprietary steps and updated procedure with UCERD information from syssccn.doc in order to eliminate it (duplicate procedure).